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AI-Driven Hybrid Fleet Optimization for Fracturing Operations: Integrating Idle Management and Electrification for Net Zero Gains

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Abstract

This paper explores a hybrid approach to decarbonizing unconventional fracturing operations by combining diesel-powered frac units equipped with Artificial Intelligence (AI)-based idle management and partially electrified dual-pump units. The objective is to demonstrate how intelligent fleet control can optimize energy use, reduce emissions, and enhance operational efficiency. The integration of AI and electrification provides a scalable framework for operators to progress towards Net Zero while maintaining performance and reliability. The hybrid system comprises electrified pump units powered by dedicated mobile natural gas gensets running on nearby pipeline gas and diesel units enhanced with idle management. AI-based intelligent control systems monitor power demand, optimize unit engagement, and transition between electric and diesel operations in real time. The control strategy incorporates predictive algorithms, cold-start optimization, and cloud-connected dashboards for live asset tracking and emissions monitoring. Combined with a cloud solution, seamless data processing, visualization, and KPIA-based decision-making aligned with standardized carbon footprint tracking methodologies are enabled.

The integrated hybrid fleet, demonstrated in 2023 in Texas, United States of America, significant environmental and operational benefits. Electrified dual pump units (7,000 hydraulic horsepower (hhp) combined) displaced substantial diesel usage, while AI-enabled idle management reduced diesel engine idle time by up to 90%. Carbon Dioxide Equivalent (CO_2e) emissions were cut by approximately 4,000 metric tons annually per frac spread. Diesel displacement via idle reduction resulted in over 2.3 million liters of fuel saved per year. The lifecycle of diesel units was extended by over two years due to reduced operational strain, and the need for auxili ary starting equipment was eliminated. Advanced data analytics verified these performance improvements under Carbon Footprint Tracking Standards. Real-time monitoring allowed adaptive fleet behavior, ensuring optimal load balancing across units and preventing overcapacity or energy waste. This automated orchestration of mixed power sources increases redundancy and ensures operational continuity. The hybrid control concept also proved scalable across multiple fleets and compatible with existing infrastructure. These results offer a replicable framework for upstream operators aiming to decarbonize high-load hydraulic fracturing operations without compromising execution efficiency or safety.

This paper introduces an AI-enhanced hybrid fleet concept that unifies idle management, electrification, and intelligent orchestration in fracturing operations. The integration of cloud analytics, predictive

automation, and dynamic control algorithms offers a pioneering model for Net Zero alignment. This approach enables performance gains and emissions reductions beyond the sum of individual technologies, setting a new benchmark for sustainable and data-driven field operations.

Introduction

Unconventional fracking operations place high demands on energy supply, availability, and emissions control. While fracking is mainly used to extract oil, it also produces a significant amount of natural gas as a by-product. Instead of flaring or releasing this unused gas into the atmosphere, the hybrid fleet concept focuses on its targeted use for energy.

Recent surveys emphasize the digital transformation of hydraulic fracturing under the Industry 4.0 paradigm, highlighting the potential of real-time parameter optimization, intelligent allocation of fracturing fluids, and collaborative multi-machine control strategies [1]. These findings underline the relevance of integrating AI-based control into frac fleets to ensure both efficiency and sustainability.

The hybrid fleet described here is based on partial electrification: not all units are fully electrified, but only part of the main drives and selected functions, in order to achieve maximum technical efficiency while maintaining a high degree of flexibility.

This partial electrification requires a high degree of automation in order to reap the benefits in terms of efficiency.

The core innovation lies in intelligent AI control with adaptive operating point selection, supported by a local smart grid and a cloud- based data platform. The aim is to optimize the energy efficiency of the fleet, significantly reduce emissions, and lower operating costs in the long term.

The following article explains the system architecture, the methods of AI-based characteristic map detection of synthetic fuel efficiency maps, and the operational advantages based on quantifiable key figures.

The objective of this paper is therefore to evaluate the feasibility and benefits of integrating AI-driven idle management with partial electrification of frac fleets. The study addresses the following research questions: (1) Can AI control strategies significantly reduce emissions and fuel consumption under field conditions? (2) How does hybridization compare to conventional diesel or fully electric fleets in terms of efficiency, lifecycle extension, and scalability?

Literature Review

To position this work within the state of the art, previous studies on ML (ML) and AI in fracturing operations are reviewed. The following contributions are particularly relevant for the hybrid fleet concept.

The application of AI and ML in hydraulic fracturing has been increasingly studied in recent years, providing a foundation for the hybrid fleet concept introduced in this paper.

A systematic overview of novel ML techniques applied to hydraulic fracturing optimization is provided in [2], covering hyperparameter tuning, surrogate-based algorithms, and real-time monitoring approaches. Their work establishes a broad methodological context for the adaptive control strategies presented here.

A large-scale ML framework on more than 16,000 field datapoints using Random Forest, Support Vector Machines, and Neural Networks is developed and validated in [3]. Their model achieved an R^2 of 0.9804 and low mean absolute error and root mean square error, demonstrating the accuracy and scalability of ML methods for predicting fracturing efficiency.

In a complementary study [4], a comparison between Artificial Neural Networks (ANN), Random Forest (RF), and K-Nearest Neighbors (KNN) algorithms for predicting fracture width under varying geological and operational parameters was carried out. Random Forest yielded the best performance with a root mean square error of 6.194, reinforcing the suitability of ensemble methods for operational decision-making.

Model-based system analysis for smart grid operations in energy-intensive environments has been demonstrated in [5], highlighting the value of predictive load management and adaptive control loops for emission reduction.

These references demonstrate the maturity of AI and ML methods for hydraulic fracturing optimization. However, little has been published on their integration with hybridized fleets combining diesel and electrified units. This paper aims to address this gap.

System architecture of the hybrid fleet

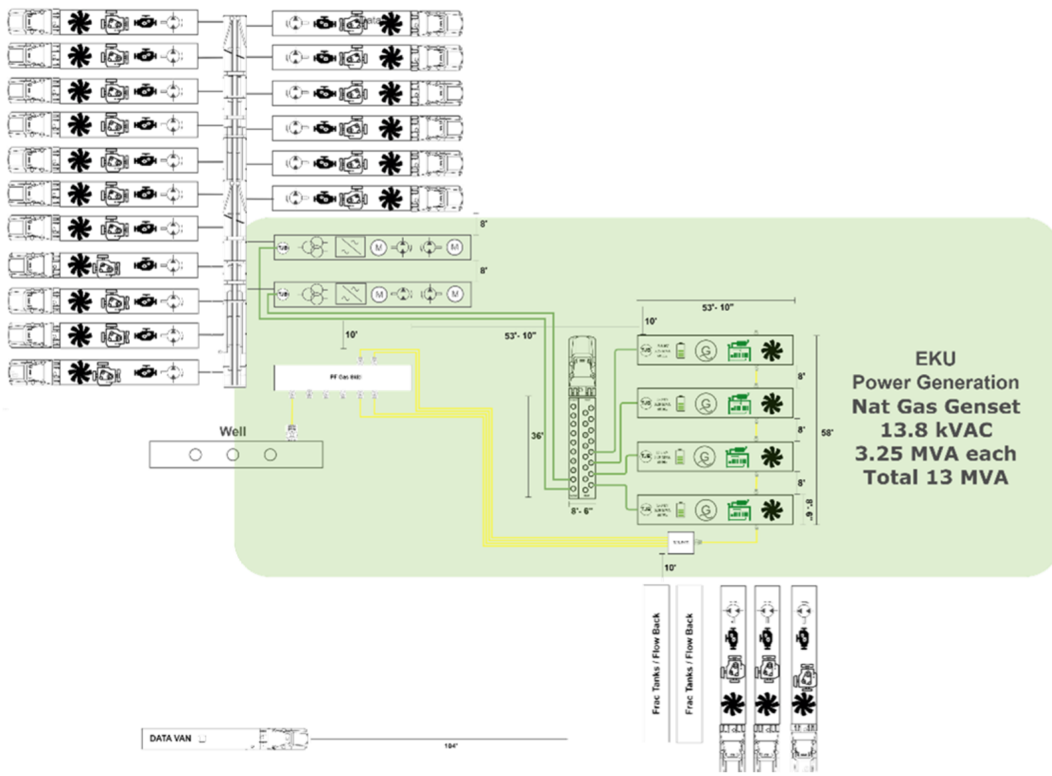

Figure 1—Layout of Hybrid Frac Spread

Composition of the fleet

The hybrid fleet typically consists of 12 to 20 classic diesel-powered pump units, supplemented by electric pump units, power generators, and, if necessary, electrified auxiliary units such as sand mixers and mixing systems. The setup includes the following components:

- Diesel-powered fracking units equipped with idle management
- Electrified fracking units: 2x dual pumps, each with 2x 3000 kW electric pumping capacity
- Electrified auxiliary units
- Gas-powered generators (Gensets) for generating electricity from frac by-product gas

The power ratings are up to 20,000-30,000 hhp per spread, with typically 16-24 units active at any one time.

Electric fracking pumps

This setup uses two so-called dual pumps, meaning that one unit contains two pumps. It is a fully electric, dual-powered high-pressure pump system for fracking applications. It has two electric drives, each with 2,700 hp, and achieves a flow rate of up to 2x 1,600 l/min at a pressure of 1,035 bar. The system offers automated warm-up, dual lubrication, real-time insulation monitoring, cloud access, precise pressure and flow control, and a compact design. The operating voltage is 1,000 V, supplemented by a 6 kWh battery system. Operating temperature range: -30 °C to +50 °C. The unit measures 16.2 m in length and weighs 73 t.

Two of these systems are used in the setup described.

Diesel-powered fracking pumps with idle management

The diesel generators are classic diesel-powered fracking pumps. The trailer-mounted fracking pump unit is designed for high-pressure use in hydraulic downhole treatment. It is based on a Cummins QSK50 diesel engine with 2,250 hp at 1,900 rpm, emission-certified and with a hydraulic start system. Power is transmitted via an 8-speed automatic transmission (Twin Disc TA90-8501) with torque converter and external oil cooling.

The installed C-2500 Quintuplex plunger pump (8" stroke) delivers up to 21.13 BPM flow rate at max. 330 rpm, depending on plunger diameter and gear selection. Two 3" 1502F flanges serve as high-pressure outlets.

A hydraulically driven cooling system with separate circuits for the engine, pump and hydraulic system ensures thermal stability. It is controlled by a local touchscreen control panel with safety features such as automatic pump shutdown in the event of overpressure.

The unit measures 13.68 m long, 3.20 m high and 2.57 m wide and is designed for use on remote drill fields. The main advantages are mobility, high performance and robust design – with typical diesel consumption and emissions.

The systems are equipped with an Engine Standby Controller (ESC). This is an idle management system. It replaces traditional hydraulic start systems and enables fully automated start/stop logic that is integrated into the AI control system of the overall system. The controller detects idle times, which occur during the conversion of the system, for example, and shuts down the motor. These are thus reduced by up to 98%. The engine is kept in standby mode, which means that all ancillary equipment, such as lubrication, cooling and heating of the individual machine, remains active, so that the engine can be restarted within a second when the power is to be retrieved. As an option, the system can be expanded to include a diesel parking heater, which keeps the system at temperature in cold regions and thus enables the start time of one second even at -40°.

In this setup, 17 of these systems are used.

Control

The control system is basically decentralized.

Each of the two electric fracking pumps, as well as the diesel gensets and the idle management systems, have their own contro l units, so that the pumps are working independently and semi-autonomous in certain functions. The start-stop functionality is therefore also available locally and stand-alone. This increases the resilience of the system. All components are networked with each other and transmit the data to the central control unit via a hybrid-cloud-based platform.

The actual fleet intelligence results from the sum of the individual controllers with the corresponding actuators but is implemented by means of a Maximus Multi-Pump Controller. This is a central controller for fleet automation of multi-motor applications. It includes the AI functions of the setup. The operating parameters of the individual units are evaluated across the board. In this way, the map is recorded and optimized.

Microgrid topology

The electrified pump units are connected via a decentralized 13,8 kV AC microgrid. In this case, the microgrid is created from 4 gas generators, each with an output of 3.25 MVA. The use of a microgrid enables the exchange of energy between the aggregates and at the same time ensures redundancy. The connection is made via mobile, easily configurable plug connections. A central control system is not required; instead, all units act cooperatively via networked control loops.

Electrified auxilliaries

Ancillary units can also be electrified in this setup. By integrating electric fans, lubricating oil pumps and auxiliary systems based on a 480 V AC link, hydraulically driven auxiliary units are replaced. The electrical energy comes from generators that run on pipeline gas. This enables more efficient partial load control and reduces mechanical wear. The electrified auxiliary units are also integrated into the control system. However, the control effort here is limited to predicting energy and power consumption with a corresponding reaction on the part of the microgrid.

Adaptive map detection with AI

Fracking fleets operate in dynamic environments with highly volatile load profiles, irregular service lives and constantly changing operating conditions. In such scenarios, manual optimization of engine operation is neither reliable nor efficient. The aim of AI-basedmap recognition is therefore to create an automated, adaptive decision-making unit that selects the best operating point in real time in terms of energy and operation.

Previous control concepts rely on predefined static characteristic curves or operator intelligence. The latter often leads to suboptimal behavior – for example, due to unnecessary revving, inefficient half-load behavior or delayed shutdown. Adaptive map recognition replaces these procedures with continuously learning logic that is tailored to the actual operating data. This significantly reduces consumption, emissions and wear.

Figure 2 and Figure 3 show typical characteristic curves, each for a diesel-powered and an electrically driven fracking pump.

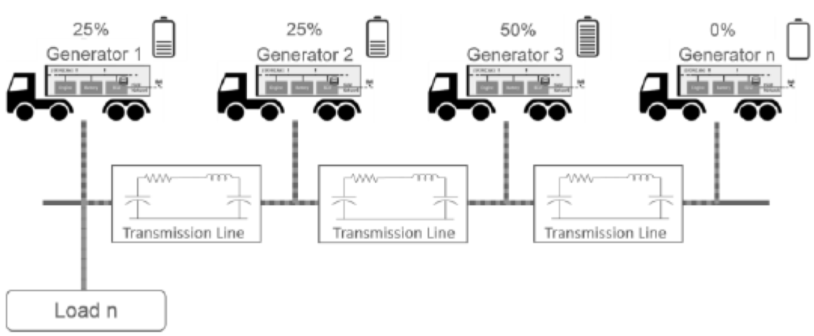

Figure 2—Load sharing in a microgrid

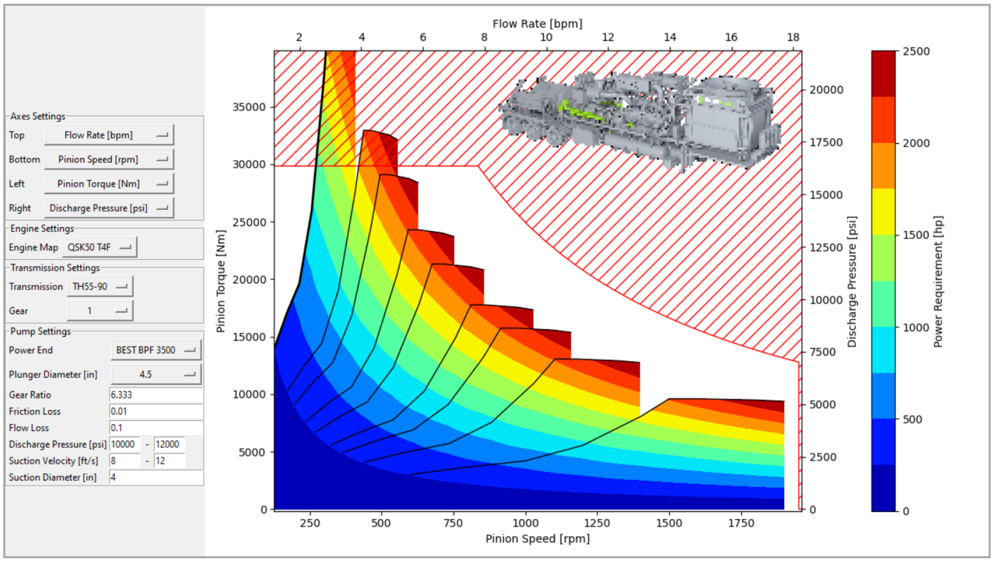

Figure 3—Diesel Pump Asset Performance Chart

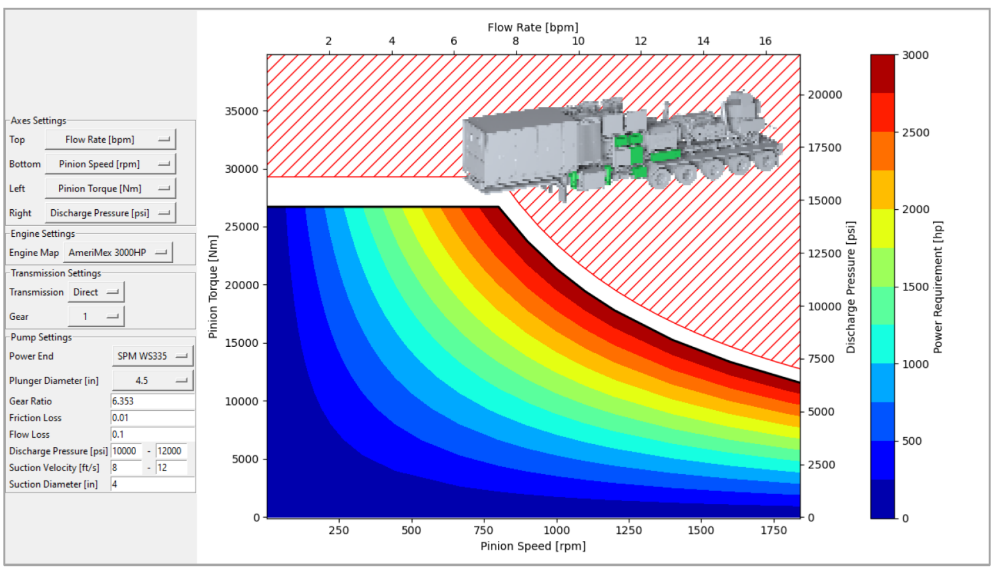

Figure 4—Performance Chart of Electric Dual Pump Unit (per Pump)

ML Model

The ML model is an individual model for each frac setup. It is created by combining the different participants (pump model, diesel pump model, electric pump model and genset model) of the frac setup.

The ML model for map recognition is integrated directly into the individual electronic control units (ECUs), such as that of the Engine Standby Controller (ESC). It is based on supervised learning methods that work with multivariate input data such as pump speed (n), outlet pressure (p), coolant temperature (T), torque (M) and energy demand (E). The target is the specific fuel consumption [g/kWh].

The operational map of the fracturing system is modeled as a multidimensional operating point map, capturing key energetic and mechanical parameters such as pump speed, outlet pressure, fuel consumption, and torque. This map forms the foundation for all data- driven control decisions during operation.

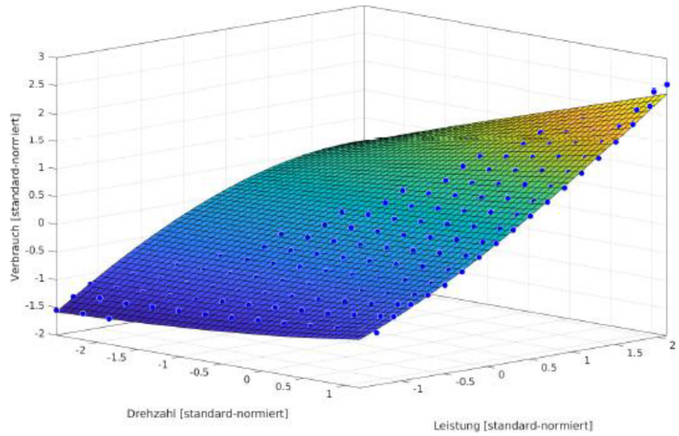

Figure 5—Operational map of a Diesel engine

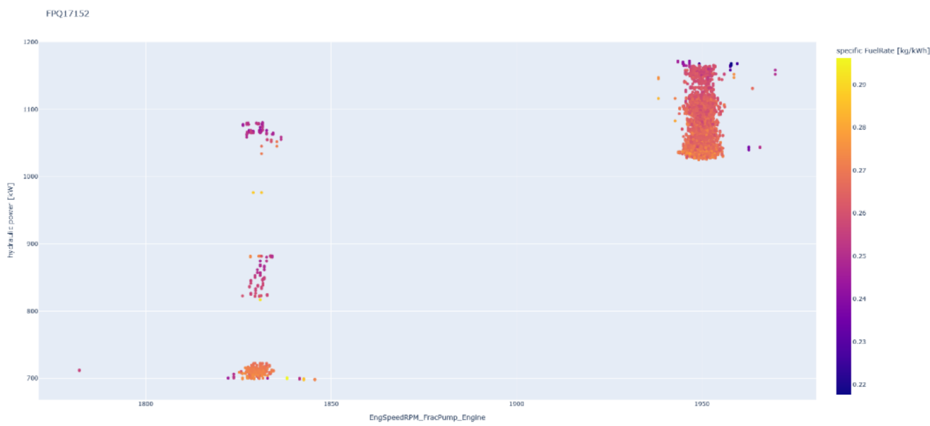

Figure 6—Operational map of a diesel driven frac pump

For practical implementation, the model uses a grid-based resolution of 100 * 100 discrete operating points, resulting in a total of 10,000 unique combinations. Each point on the map represents a specific configuration of process variables and is linked to measured or predicted performance values. This fine-granular structure allows the system to accurately reflect both stationary and transient behavior across the full operating range.

The initial training phase of the ML model was conducted offline using historical operating data from real frac operations. These data were grouped into representative operational clusters, enabling robust generalization and improved model convergence. The clustering included Start-up phases, where thermal and mechanical dynamics are most pronounced, Idle phases, focusing on minimizing fuel use and optimizing standby performance, Constant load operation, typically targeted for energy efficiency, Partial load scenarios, which often suffer from inefficiency and offer high optimization potential and High-load states with transients, such as pressure fluctuations or fast load changes.

This structured preprocessing allows the learning algorithm to capture condition-specific patterns and transition behaviors. The chosen clustering method follows insights from model-based system analysis [5], demonstrating that separating these regimes improves training stability and yields more robust predictions in live environments.

By combining clustered offline training with continuous online adaptation, the result is a high-resolution, context-sensitive operation map that not only maximizes fuel efficiency but also accounts for wear, system responsiveness, and emissions as part of the real-time decision logic.

Continuous online learning takes place during real-time operation. The model is dynamically updated to compensate for drift effects (e.g. due to component aging, weather or load shifts). The weighting is done via a Bayesian update procedure that links priority distributions with new data.

Online adaptation in the field

The AI system works completely autonomously and locally. Each pump unit learns from its own operating history. The operating parameters are continuously evaluated via the integrated cloud platform, and an overarching strategy specification is issued. For example, the desired total pump flow is specified at total pressure. In addition, the strategy specification can be expanded to include, for example, optimization for maximum efficiency, minimum emissions or longest service life. The local ESC unit receives this input and adapts the prioritization logic accordingly.

The online adaptation reacts to fluctuations such as, Water temperature (influence on cooling limits), Time of day (start/stop behavior at changing temperatures), Frac stage (pressure build-up, peak flow, holding phase) and Aggregate state (availability assessment via digital twin).

Like intelligent cruise control in the automotive sector, AI regulates the optimal pump speed and the corresponding pressure range in such a way that fuel consumption is within the optimal range. There is no need for faulty operator intervention or delayed reactions. The result: +12% efficiency, –18% emissions (measured in pilot projects with 4 * ESC pumps cumulative over 1,600 h).

Architecture

The control hardware of the individual actors is based on an ARM Cortex-M7 with a dedicated FPU (Floating Point Unit). A RAM buffer of 512 kB is available for the real-time model, which is queried cyclically by the ESC rules software. The ML model is modular and can be updated by OTA update via SOPHIA.

Communication is done via MQTT with SOPHIA, internally via J1939-CAN. In the event of a communication failure, the ML system continues to operate locally. For fleet operation, the architecture is designed in such a way that learning effects from other units can also be aggregated ("transfer learning").

Building on the system architecture described, the following section outlines the methodology applied to evaluate hybrid fleet performance. The methodology distinguishes between system-level design, AI-based control approaches, and emission quantification methods.

These principles form the foundation of the methodology described in Section 4, where AI-based control strategies are combined with deterministic safety layers

Methodology and control strategy

Hybrid Control: Combination of AI and Deterministic Control Algorithms

The control architecture of the hybrid frac setup is based on a multi-level control approach that combines an adaptive AI component with hard-coded safety and system boundaries. The AI, implemented in the form of an edge-based ML module on each unit, continuously calculates the energetically optimal state of the system based on the current operating point, historical data and target parameters (e.g. pressure, flow, CO_2 targets). The results of the AI are transferred as a tax specification to the set of rules, which act within previously defined limits.

In concrete terms, this means that while the AI calculates an optimal target value for speed and pressure, a parallel deterministic block checks whether this target value is compatible with safety-relevant operating conditions. Typical conditions include exceeding an oil temperature of 95 °C results in activating speed limits, Pressure increase of more than 10 bar/s results in activation of a ramp to avoid hydraulic shocks and Battery level below 30% results in automatically requesting additional power by starting a genset.

This principle of an "AI-in-the-loop" structure allows data-driven optimization without compromising security and operational limits. The intelligent control system is based on the controls platform, which also supports functions such as automatic ramp-up profiles, pump replacement in the event of failure and prediction of load profiles. The aim is stable and at the same time adaptive operation under dynamic field conditions.

The special feature of the control principle used is its continuous adaptation: As soon as framework parameters change – for example due to weather influences, system aging or deviating media behavior – the AI continuously adapts the map structure. At the same time, the deterministic protection circuits operate at a predefined limit level. This decoupling allows both efficiency gains to be realized and compliance requirements to be met.

Decentralized multi-agent regulation

A distributed multi-agent approach is used to orchestrate the electric and hybrid units. Each agent – usually a pump unit or generator – has an autonomous control instance with local control algorithms. These instances coordinate with each other via a decentralized communication protocol based on the Real-Time Publish/Subscribe (RTPS) standard, which is part of the Data Distribution Service (DDS).

Each agent includes A primary control system for direct voltage and current control, A secondary control for fine adjustment based on ΔV (voltage deviation) and AI (current deviation) and Observer functions for global average estimation of network sizes

The agents communicate their local assessments of the network parameters with each other. Iterative consensus algorithms, based on a Laplace matrix of the communication graph, create a stable overall state without the need for a central host computer. The advantages of this architecture are No single point of failure, Arbitrary scaling of system size and Robustness against grid failures or topology changes

Real-time emission control

The integration of the cloud platform allows for higher-level, emission-based control. CO_2e. NO_X emissions and other parameters are continuously measured and evaluated using KPIs. The control targets are specified as operational target values ("operate below 250 g CO_2/kWh") and are adopted in real time in the control of the individual units.

The AI's decision matrix simultaneously considers fuel consumption, emission characteristics and dynamic load and wear indicators.

These criteria are weighted by predefined priority profiles or operator-side entries. On this basis, the cloud platform automatically generates reports in XML, PDF or WITSML format, which can be used for Environmental, Social, and Governance (ESG) reports, customer communication or internal optimization. The combination of adaptive emissions control and dynamic fleet orchestration ensures that regulatory targets are met while exploiting technical efficiency potential.

Beyond operational optimization, emission quantification is a key requirement for regulatory compliance and ESG reporting. A recent design-driven tool allows accurate CO_2e footprint assessments across all job phases. enabling a more robust evaluation of hy brid fleet performance (SPE 221941, 2025). Incorporating such methods ensures that reported emission reductions are traceable and comparable across different operations.

Results and key figures

Emission reduction

In pilot projects with fleets of two electric a 17 diesel units (all equipped with ESC and AI map control). an average CO_2e reduction of 3,950 t/year was achieved. This corresponds to approximately 20% savings compared to conventional frac spreads. The frac gas, which would otherwise have been flared, could be used for energy purposes to about 85%. This also results in a reduction in methane emissions (by avoiding flaring) of about 8-12%.

Table 1—Results

Metric	Baseline	Hybrid Fleet	Improvement
CO_2e Emissions	~20,000 t/y	16,050 t/y	–20%
Diesel consumption	11.5 Mio L/y	9.1 Mio L/y	–2.36 Mio L
Idle Time	100%	7%	–93%
Cold Starts	100%	22%	–78%

These findings are in line with broader industry observations. Field trials reported by Oil & Gas Journal (2025) demonstrated that AI- driven fracturing operations reduced non-productive stages by up to 78% and improved proppant placement efficiency. Such independent evidence strengthens the interpretation of the presented results.

Comparable reductions were also reported by Akbari et al. (2025), where ML-based control strategies improved overall efficiency under variable load conditions. These independent findings underline the robustness of AI-enhanced operation.

Operational Efficiency

The no-load reduction was up to 93%. The average load factor increased to 49 % for ESC-equipped units (base: 800 h measurement period), compared to 24 % without ESC. The diesel savings amounted to approx. 2.36 million liters per spread and year. The grid utilization of the generators was also significantly stabilized.

Similar improvements in operational utilization factors were highlighted by Karami et al. (2025), who demonstrated the advantage of Random Forest models in predicting fracture behavior under varying geological conditions. This supports the conclusion that AI-driven predictive control enhances both energy efficiency and stability.

Lifespan and Maintenance

The number of cold starts has been reduced by an average of 78%. This extended the service life of individual diesel generators by up to two years. At the same time, maintenance costs were reduced due to the elimination of external starting systems and less wear and tear on auxiliary units. In addition, the oil change frequency dropped significantly.

System resilience

The decentralized control system showed a very high level of robustness in simulations and real failures (e.g. failure of a pump controller). Response times during load changes were on average < 300 ms. Diesel gensets successfully compensated for the sluggish reactions of the gensets. The system demonstrated high reliability in the event of dynamic topology changes.

Discussion and outlook

The presented solution combines several levels of innovation like Automated AI operating point selection for optimal efficiency, Partial electrification for maximum feasibility, Decentralized control via multi-agent architecture and Use of frac gas as a valuable energy source.

These results confirm and extend the findings of recent reviews (Alake et al., 2024; Jia et al., 2024) that emphasized the transformative role of AI in hydraulic fracturing. Unlike previous work, this study demonstrates how adaptive ML approaches can be embedded directly into operational control loops of hybrid fleets, ensuring practical applicability at scale.

Compared to purely electric fleets, the concept avoids the infrastructural hurdles (e.g. lack of grid connection), while at the same time being much lower in emissions than conventional diesel spreads. The combination of AI-controlled fine control and robust automation concept raises the operating quality to a new level. It is particularly noteworthy that the system can also react to long-term changes (e.g. aging, relocation, weather) through online learning.

The literature confirms that the combination of AI-based optimization and hybridization bridges the gap between conventional diesel fleets and fully electric operations. As highlighted by Jia et al. (2024) and Akbari et al. (2025), adaptive ML approaches are capable of delivering stable efficiency improvements even under volatile load conditions. Our results align with these findings, extending them by demonstrating large-scale applicability in real frac spreads.

Major challenges remain in adaptation to large fleets (>20 units), synchronization over long cable connections, and interoperability with legacy systems. However, the already high modularity of the platform allows for an incremental rollout concept.

Inference

The hybrid concept presented by EKU Power Drives shows how a realistic, economically viable and ecologically sensible transformation of unconventional frac fleets can be achieved through intelligent subsystem electrification, adaptive AI control and the use of existing resources (e.g. by-product gas).

Outlook

Future developments in hybrid fleet architecture will focus in particular on two areas of expansion:

Firstly, the integration of stationary battery storage systems on the well site is sought. By using high-performance lithium-ion or solid- state batteries, short-term load peaks can be absorbed, and generators can be operated in the optimal load range. This increases the overall efficiency of the drive sources and reduces both specific emissions and fuel consumption. At the same time, storage integration improves grid stability and allows additional functionalities such as peak shaving and grid support – even in temporary or island-based supply structures.

Secondly, the focus is on integration into higher-level energy management systems (EMS). The hybrid fleet is not only optimized locally but also controlled as part of a larger grid network (e.g. drilling field, region, supply chain). Dispatch decisions are made on the basis of CO2 intensity, real-time consumption data, TCO key figures and regulatory frameworks. The aim is to dynamically adapt the hybrid fleet to the most economically and ecologically sensible operating strategy. The combination of local learning, fleet intelligence and higher-level strategy management opens up new potential for grid-connected, resilient and climate-optimized frac operations of the next generation.

Conclusions

This paper demonstrated that AI-driven idle management combined with partial electrification can achieve substantial efficiency gains and emission reductions in fracturing operations. Key findings include:

- up to 20% annual CO2e reduction per spread,
- diesel fuel savings exceeding 2.3 million liters per year,
- extension of diesel unit lifespan by up to two years.

The hybrid fleet approach avoids the infrastructural limitations of full electrification while offering a scalable, modular solution. Future research should focus on integrating battery storage and higher-level EMS systems to further enhance resilience and sustainability.

Beyond operational efficiency, the presented approach supports ESG compliance and offers a scalable pathway for upstream operators to progress toward Net Zero without full electrification. Future work will need to validate long-term economic impacts, especially under varying gas availability and regulatory frameworks.

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